

Realtek Wi-Fi SDK for Android KK 4.4

ver. 1.1.0

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Release History

0.0.1	2013/11/22	1. Beta release 1.1. realtek_wifi_SDK_for_android_KK_4.4_20131122.tar.gz 1.2. wpa_supplicant_8_kk_4.4_rtw_r9640.20131122.tar.gz
1.0.0	2014/01/17	1. First formal release 1.1. realtek_wifi_SDK_for_android_KK_4.4_20140117.tar.gz
1.1.0	2014/02/20	1. wpa_supplicant_8_kk_4.4_rtw_r10450.20140220.tar.gz

SDK packages

- hardware/realtek/*
Folder to store config files, private code from Realtek.
- hardware/libhardware_legacy/wifi/Android.mk
Reference codes for applying wifi_realtek.c

※ For wpa_supplicant_8_kk_4.4_rtw_r10450.20140220.tar.gz or newer version, see wpa_supplicant_hostapd folder of our SW release package or consult our contact window.

Introduction

This document provides a simple guide to help engineers to apply Realtek Wi-Fi solution onto their Android KK 4.4 system. For now, we have supported the following two scenarios:

- **STA/AP** Switch between STA mode and AP mode
- **(STA+P2P)/AP** Switch between STA+P2P(Wi-Fi Direct) concurrent mode and AP mode

To port Realtek Wi-Fi driver onto Android 4.4 platform, you can go through the following guide with reference codes within our driver package's realtek_wifi_SDK_for_android_KK_4.4_20140117.tar.gz.

Because Android's SDK may differ from platform to platform, our reference codes may not be applied on every platform without modifications. You should check if our reference code is suitable for you to use.

In this document, ANDROID_SDK is the path of top folder of the target Android SDK; this term is used in the following paragraphs.

1. Copy Necessary Files into SDK

After unzipping `realtek_wifi_SDK_for_android_KK_4.4_20140117.tar.gz`, copy the following folder into `ANDROID_SDK/hardware/` folder:

- `hardware/realtek`

2. Platform Related Files

2.1. BoardConfig.mk

To apply Realtek Wi-Fi solution onto your Android KK system, you need to define some compile-time variables in `BoardConfig.mk` of your platform. In general, the `BoardConfig.mk` file is located in:

`ANDROID_SDK /device/<soc_vendor_name>/<board_name>/`

Take TI panda board for example:

`ANDROID_SDK /device/ ti/panda/ BoardConfig.mk`

Please define the following compile-time variables below:

```

BOARD_WIFI_VENDOR := realtek
ifeq ($(BOARD_WIFI_VENDOR), realtek)
    WPA_SUPPLICANT_VERSION := VER_0_8_X
    BOARD_WPA_SUPPLICANT_DRIVER := NL80211
    CONFIG_DRIVER_WEXT :=y
    BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl
    BOARD_HOSTAPD_DRIVER      := NL80211
    BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl

    BOARD_WLAN_DEVICE := rtl8192cu
    #BOARD_WLAN_DEVICE := rtl8192du
    #BOARD_WLAN_DEVICE := rtl8192ce
    #BOARD_WLAN_DEVICE := rtl8192de
    #BOARD_WLAN_DEVICE := rtl8723as
    #BOARD_WLAN_DEVICE := rtl8723au
    #BOARD_WLAN_DEVICE := rtl8189es
    #BOARD_WLAN_DEVICE := rtl8723bs
    #BOARD_WLAN_DEVICE := rtl8723bu

    WIFI_DRIVER_MODULE_NAME  := "wlan"
    WIFI_DRIVER_MODULE_PATH  := "/system/lib/modules/wlan.ko"
    WIFI_DRIVER_MODULE_ARG   := "ifname=wlan0 if2name=p2p0"

    WIFI_FIRMWARE_LOADER     := ""
    WIFI_DRIVER_FW_PATH_STA  := ""
    WIFI_DRIVER_FW_PATH_AP   := ""
    WIFI_DRIVER_FW_PATH_P2P  := ""
    WIFI_DRIVER_FW_PATH_PARAM := ""
endif

```

- **BOARD_WIFI_VENDOR := realtek**

To distinguish the platform Wi-Fi device from products of other vendors, we define variable BOARD_WIFI_VENDOR as realtek. This is for compile-time choices to be applied for Realtek Wi-Fi solutions.

- **WPA_SUPPLICANT_VERSION := VER_0_8_X**

For Android KK, please set WPA_SUPPLICANT_VERSION as VER_0_8_X to

use wpa_supplicant_8.

- **BOARD_WPA_SUPPLICANT_DRIVER := NL80211**
- **BOARD_WPA_SUPPLICANT_PRIVATE_LIB := lib_driver_cmd_rtl**
- **BOARD_HOSTAPD_DRIVER := NL80211**
- **BOARD_HOSTAPD_PRIVATE_LIB := lib_driver_cmd_rtl**

We use NL80211 as the driver interface for wpa_supplicant and hostapd to communicate with driver and provide lib_driver_cmd_rtl as the private library.

- **BOARD_WLAN_DEVICE**

Realtek provide a variety of Wi-Fi solutions to choose. For now, BOARD_WLAN_DEVICE is not used for any purpose but we suggest setting this variable for your Wi-Fi solution you used.

- **WIFI_DRIVER_MODULE_NAME**
- **WIFI_DRIVER_MODULE_PATH**
- **WIFI_DRIVER_MODULE_ARG**

These three variables will be used in libhardware_legacy (wifi.c/wifi_realtek.c) to do insmod and remmod. The value of WIFI_DRIVER_MODULE_NAME should match the value of MODULE_NAME specified in our driver's Makefile at compile-time. Please refer to Platform Setting Section in Detail of:

document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf

- **WIFI_FIRMWARE_LOADER :=""**
- **WIFI_DRIVER_FW_PATH_STA :=""**
- **WIFI_DRIVER_FW_PATH_AP :=""**
- **WIFI_DRIVER_FW_PATH_P2P :=""**
- **WIFI_DRIVER_FW_PATH_PARAM :=""**

Because our driver has FW embedded inside, and will automatically load FW at NIC initialization process, there is no need to set these 5 variables, just keep them empty.

2.2. init.xxx.rc

For Wi-Fi to operate properly, we need some daemons to be defined as service inside init.xxx.rc. In general, the init.xxx.rc file is located in:

ANDROID_SDK/device/<soc_vendor_name>/<board_name>/

Take TI panda board for example:

ANDROID_SDK/device/ti/panda/init.omap4pandaboard.rc.

Please add the service definitions below:

- **wpa_supplicant**

```
service rtw_suppl_con /system/bin/wpa_supplicant \
    -ip2p0 -Dnl80211 -c/data/misc/wifi/p2p_supplicant.conf \
    -e/data/misc/wifi/entropy.bin -N \
    -iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \
    -O/data/misc/wifi/sockets \
    -g@android:wpa_wlan0
class main
socket wpa_wlan0 dgram 660 wifi wifi
disabled
oneshot

service rtw_suppl /system/bin/wpa_supplicant \
    -iwlan0 -Dnl80211 -c/data/misc/wifi/wpa_supplicant.conf \
    -O/data/misc/wifi/sockets \
    -e/data/misc/wifi/entropy.bin
    -g@android:wpa_wlan0
class main
socket wpa_wlan0 dgram 660 wifi wifi
disabled
oneshot
```

- **dhcpcd**

```
service dhcpcd_wlan0 /system/bin/dhcpcd -aABDKL
    class main
    disabled
    oneshot

service dhcpcd_p2p /system/bin/dhcpcd -aABKL
    class main
    disabled
    oneshot

service iprenw_wlan0 /system/bin/dhcpcd -n
    class main
    disabled
    oneshot

service iprenw_p2p /system/bin/dhcpcd -n
    class main
    disabled
    oneshot
```

2.3. Others

For topics mentioned here, you can add the following code segments in any .mk file which your platform will use. Take TI panda board for example:

ANDROID_SDK /device/ ti/panda/device.mk.

- **Add android.hardware.wifi.xml**

To claim Wi-Fi support for you device, please add the rule in the PRODUCT_COPY_FILES variable to copy the permission definition file of Wi-Fi to the /system/etc/permissions/ folder of your system image.

```
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.wifi.xml:system/etc/permissions/android.hardware.
wifi.xml
```

- **Add android.hardware.wifi.direct.xml**

To claim Wi-Fi Direct (P2P) support for your device, please add the rule in the

- **radioAttributes**

To define the system's available network interfaces, we need to define interface items for wifi in the radioAttributes resource configuration. For example:

```
<string-array translatable="false" name="radioAttributes">  
    <item>"1,1"</item>  
    <item>"7,1"</item>  
    <item>"9,1"</item>  
</string-array>
```

- **config_tether_wifi_regexs**

The interfaces set here are tetherable Wi-Fi interfaces which will be used as interfaces for Wi-Fi LAN port. We use 'wlan0' by default when our Wi-Fi is set as softap mode. So it needs to set 'wlan0' here. For example:

```
<string-array translatable="false" name="config_tether_wifi_regexs">  
    <item>"wlan0"</item>  
</string-array>
```

- **config_tether_upstream_types**

The connection types set here are used as the interfaces for WAN port to connect to internet. For example, adding Wi-Fi and Ethernet:

```
<integer-array translatable="false" name="config_tether_upstream_types">  
    <item>1</item>  
    <item>9</item>  
</integer-array>
```

At least one item should be declared here to enable the Tethering&portable hotspot option of WirelessSettings in Settings.apk.

To know the definition and set other upstream connection types, please refer to ANDROID_SDK/frameworks/base/core/java/android/net/ConnectivityManager.java.

4. **libhardware_legacy**

The libhardware_legacy library includes functionality for Wi-Fi to operate. We have made modifications and extensions for our Wi-Fi solutions. To apply this, please modify ANDROID_SDK/hardware/libhardware_legacy/wifi/Android.mk to include

wifi_realtek.c instead of wifi.c into LOCAL_SRC_FILES. For example:

```
ifeq ($(BOARD_WIFI_VENDOR), realtek)
LOCAL_SRC_FILES += ../realtek/wlan/libhardware_legacy/wifi/wifi_realtek.c
else
LOCAL_SRC_FILES += wifi/wifi.c
endif
```

5. wpa_supplicant_8

We provide wpa_supplicant_8_kk_4.4_rtw_r10450.20140220.tar.gz or newer version in the wpa_supplicant_hostapd/ of our SW release package. You can:

- **Compare and merge with your own wpa_supplicant_8**

Compare and merge from wpa_supplicant_8_kk_4.4_rtw by your own. For both ANDROID_SDK/external/wpa_supplicant_8/wpa_supplicant/Android.mk and ANDROID_SDK/external/wpa_supplicant_8/hostapd/Android.mk, you should notice that the two macros REALTEK_WIFI_VENDOR and ANDROID_P2P should be added into L_CFLAGS. For example:

```
ifeq ($(BOARD_WLAN_DEVICE), bcm43xx)
L_CFLAGS += -DANDROID_P2P
endif

ifeq ($(BOARD_WIFI_VENDOR), realtek)
L_CFLAGS += -DREALTEK_WIFI_VENDOR
L_CFLAGS += -DANDROID_P2P
Endif

# Use Android specific directory for control interface sockets
```

Here is the description of the specific macros:

MACRO	Description	
ANDROID_P2P	Android s wpa_supplicant_8 patch.	Must
REALTEK_WIFI_VENDOR	General purpose patch made by Realtek.	Must

- **Use the wpa_supplicant_8_kk_4.4_rtw instead of the original**

A. Backup and remove the original external/wpa_supplicant_8/ folder

- B. Extract and copy the wpa_supplicant_8_kk_4.4_rtw tar file to the external/ folder of your Android SDK.
- C. Rename wpa_supplicant_8_kk_4.4_rtw as wpa_supplicant_8.

※ We have enabled the two macros ANDROID_P2P and REALTEK_WIFI_VENDOR by default.

6. Driver Configurations for Android KK

Android KK support two scenarios for Wi-Fi solution:

- **STA/AP – Switch between STA and AP mode**
- **(STA+P2P)/AP – Switch between STA+P2P concurrent and AP mode**

The configuration of driver to fit the requirement of each scenario, see the following table:

MACRO	STA /AP	(STA+P2P)/AP	Kernel ver.
CONFIG_IOCTL_CFG80211	Defined	Defined	ver. >= 2.6.35
RTW_USE_CFG80211_STA_EVENT	Defined	Defined	ver. >= 3.2.0
CONFIG_CONCURRENT_MODE	Undefined	Defined	-
CONFIG_WFD	Don't Care	Must for Miracast	-
RTW_ENABLE_WIFI_CONTROL_FUNC	Must for platform device/driver mechanism		

- **RTW_USE_CFG80211_STA_EVENT** is used for driver to indicate new cfg80211 STA event which is required by wpa_supplicant_8 of Android 4.4. Linux kernel supports this feature after kernel 3.2. For kernel version between 3.0 and 3.2, please refer to the patch file:

linux-3.0.42_STATION_INFO_ASSOC_REQ_IES.diff

- **RTW_ENABLE_WIFI_CONTROL_FUNC** is used to register platform driver callbacks. If your platform needs those callbacks, please define this macro to register platform driver callback functions. For example, these functions include:

```
static struct platform_driver wifi_device = {
    .probe          = wifi_probe,
    .remove         = wifi_remove,
```

By default, the probe callback is used to set up Wi-Fi power and remove callback is used to close Wi-Fi power.

Please modify both the include/autoconf.h and the specific autoconf file(needed

for compound driver release) for your Wi-Fi product.

Chip type

Autoconf file to modify

For example, if you want to configure RTL8192CU-series driver (ex: RTL8188CUS, RTL8192CU) to fit the scenario of (STA+P2P)/AP, make sure the macros: `CONFIG_IOCTL_CFG80211`, `RTW_USE_CFG80211_STA_EVENT` and `CONFIG_CONCURRENT_MODE` in both `include/autoconf.h` and `autoconf_rtl8192c_usb_linux.h`(needed for compound driver release) are defined. As following:

To compile our Wi-Fi driver, please refer to the following document for more informations:

[document/Quick_Start_Guide_for_Driver_Compilation_and_Installation.pdf](#)

7.3. Wi-Fi Direct (P2P mode)

7.3.1. There is no Wi-Fi Direct UI shown?

Please refer to **Add `android.hardware.wifi.direct.xml`** in chapter 2.3. Others to enable Wi-Fi Direct functionality of Android KK.

7.3.2. Wi-Fi Direct can't scan any peer?

First, make sure you have workable Wi-Fi Direct device nearby. Make them into Wi-Fi Direct scanning state. Push **SEARCH FOR DEVICES** button also in our device and wait for a while.

If there is still no peer shown the problem is usually caused by wrong service definition of `wpa_supplicant` services. Please refer to **wpa_supplicant** in chapter 2.2. **init.xxx.rc** to check your service definition of `wpa_supplicant`.